Exercise of Supervised Learning: Advanced Risk Minimization Part 1

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Exercise 1: Risk Minimizers for Generalized L2-Loss

Consider the regression learning setting, i.e., $\mathcal{Y} = \mathbb{R}$, and assume that your loss function of interest is $L(y, f(\mathbf{x})) = (m(y) - m(f(\mathbf{x})))^2$, where: $m : \mathbb{R} \to \mathbb{R}$ is a continuous strictly monotnone function.

Disclaimer: In the following we always assume that $\text{Var}(m(Y))$ **exists.**

(a) Consider the hypothesis space of a constant models
$\mathcal{H} = \{f : \mathcal{X} \to \mathbb{R} | f(\mathbf{x}) = \boldsymbol{\theta} \; \forall \mathbf{x} \in \mathcal{X}\}$, where $\mathcal{X}$ is the feature space. Show that

$$\hat{f}(\mathbf{x}) = m^{-1}\left(\frac{1}{n}\sum_{i=1}^{n} m(y^{(i)})\right)$$

is the optimal constant model for the loss function above, where m^{-1} is the inverse function of m.

Solution to Question (a)

1. f is a **constant model**: $f(\mathbf{x}) = \boldsymbol{\theta}$ for all $\mathbf{x}$.
2. The empirical risk can be formulated as:

$$\mathcal{R}_{\text{emp}}(f) = \sum_{i=1}^{n} \left(m(y^{(i)}) - m(f(\mathbf{x}^{(i)})) \right)^2 = \sum_{i=1}^{n} \left(m(y^{(i)}) - m(\boldsymbol{\theta}) \right)^2 .$$

3. $\mathcal{R}_{\text{emp}}(f)$ is **strictly convex** (because MSE loss and m is strictly monotone). So the minimum is unique, and can be computed by solving $\partial \mathcal{R}_{\text{emp}}(f) / \partial \boldsymbol{\theta} = \mathbf{0}$.

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Solution to Question (a): Continued

Goal: Compute the optimal $\boldsymbol{\theta}$ by solving $\partial\mathcal{R}_{\text{emp}}(f)/\partial\boldsymbol{\theta} = \mathbf{0}$.

1. Compute the derivative:

$$\frac{\partial\mathcal{R}_{\text{emp}}(f)}{\partial\boldsymbol{\theta}} = 2\sum_{i=1}^{n}(m(y^{(i)}) - m(\boldsymbol{\theta})) \cdot \frac{\partial m(\boldsymbol{\theta})}{\partial\boldsymbol{\theta}} = 0$$

2. Using the fact that $\frac{\partial m(\boldsymbol{\theta})}{\partial\boldsymbol{\theta}}$ is constant for all i, we obtain:

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$$\sum_{i=1}^{n}(m(y^{(i)}) - m(\boldsymbol{\theta})) = 0$$

$$\Rightarrow \sum_{i=1}^{n} m(y^{(i)}) = \sum_{i=1}^{n} m(\boldsymbol{\theta})$$

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$$\Rightarrow m(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n} m(y^{(i)})$$

$$\Rightarrow \boldsymbol{\theta}^* = m^{-1}\left(\frac{1}{n}\sum_{i=1}^{n} m(y^{(i)})\right) \qquad \triangleleft$$

Question (b)

(b) Verify that the risk of the optimal constant model is $\mathcal{R}_L\left(\hat{f}\right) = (1 + \frac{1}{n})\text{Var}(m(y))$.

Recall that the risk of $\hat{f}$ is defined as

$$\mathcal{R}_L(\hat{f}) = \mathbb{E}_{xy}[L(y, \hat{f}(\mathbf{x})]$$

Solution to Question (b)

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\begin{aligned}
\mathcal{R}_L(\hat{f}) &= \mathbb{E}_{xy}[L(y, \hat{f}(\mathbf{x})] \\
&= \mathbb{E}_{xy}[(m(y) - m(\hat{f}(\mathbf{x}))^2]
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&= \mathbb{E}_{xy}\left[\left(m(y) - \frac{1}{n}\sum_{i=1}^{n} m(y^{(i)})\right)^2\right]
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&= \mathbb{E}_{xy}[m(y)^2] - 2 \cdot \mathbb{E}_{xy}\left[m(y) \cdot \frac{1}{n}\sum_{i=1}^{n} m(y^{(i)})\right] + \mathbb{E}_{xy}\left[\left(\frac{1}{n}\sum_{i=1}^{n} m(y^{(i)})\right)\left(\frac{1}{n}\sum_{i=1}^{n} m(y^{(i)})\right)\right]
\end{aligned}
$$

Solution to Question (b): Continued

Now take a look at the second term: $-2 \cdot \mathbb{E}_{xy}\left[m(y) \cdot \frac{1}{n}\sum_{i=1}^{n} m(y^{(i)})\right]$.

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\begin{aligned}
&\mathbb{E}_{xy}\left[m(y) \cdot \frac{1}{n}\sum_{i=1}^{n} m(y^{(i)})\right] \\
&= \frac{1}{n} \cdot \mathbb{E}_{xy}\left[m(y)\sum_{i=1}^{n} m(y^{(i)})\right] \\
&= \frac{1}{n} \cdot \mathbb{E}_{xy}[m(y)]\,\mathbb{E}_{xy}\left[\sum_{i=1}^{n} m(y^{(i)})\right]
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&= \frac{1}{n} \cdot \mathbb{E}_{xy}[m(y)] \cdot n \cdot \mathbb{E}_{xy}[m(y)]
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Why $\mathbb{E}_{xy}\left[\sum_{i=1}^{n} m(y^{(i)})\right] = n \cdot \mathbb{E}_{xy}[m(y)]$**?**

- Linearity: pull $\mathbb{E}_{xy}$ inside the sum.
- (i) just labels different draws; all $y^{(i)}$ share y's distribution, so $\mathbb{E}_{xy}[m(y^{(i)})] = \mathbb{E}_{xy}[m(y)]$.
- Sum of n equal terms $= n \cdot \mathbb{E}_{xy}[m(y)]$.

Analogy: expected sum of n dice rolls is $n \times 3.5$.

Solution to Question (b): Continued

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&= \frac{1}{n} \cdot \mathbb{E}_{xy}[m(y)] \cdot n \cdot \mathbb{E}_{xy}[m(y)] \\
&= \mathbb{E}_{xy}[m(y)]^2.
\end{aligned}
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Solution (b): third term $\mathbb{E}_{xy}\left[\left(\frac{1}{n}\sum_{i=1}^{n} m(y^{(i)})\right)^2\right]$

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$$= \frac{1}{n^2}\mathbb{E}_{xy}\left[\sum_{i,j=1}^{n} m(y^{(i)})m(y^{(j)})\right]$$

Example ($n = 3$). Let $m_i := m(y^{(i)})$.

$$\left(\frac{m_1+m_2+m_3}{3}\right)^2 = \frac{1}{9}\sum_{i,j=1}^{3} m_i m_j$$

Expand all 9 pairs (i, j):

$$\begin{array}{ccc} m_1^2 & m_1 m_2 & m_1 m_3 \\ m_2 m_1 & m_2^2 & m_2 m_3 \\ m_3 m_1 & m_3 m_2 & m_3^2 \end{array}$$

Group by color:

- Diagonal ($i = j$, n terms) $\rightsquigarrow \sum_i m_i^2$
- Off-diag. ($i \neq j$, $n(n-1)$ terms) $\rightsquigarrow \sum_i \sum_{j\neq i} m_i m_j$

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$$= \frac{1}{n^2}\Bigg(\sum_{i=1}^{n}\mathbb{E}_{xy}[m(y^{(i)})^2]$$

$$+ \sum_{i=1}^{n}\sum_{j\neq i}\mathbb{E}_{xy}[m(y^{(i)})m(y^{(j)})]\Bigg)$$

▷ Split into diagonal ($i = j$, squared) and off-diagonal ($i \neq j$, cross) terms.

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Solution (b): third term, cont.

Continuing from the grouping form:

$$\frac{1}{n^2}\Bigg(\sum_{i=1}^{n}\mathbb{E}_{xy}[m(y^{(i)})^2] \\ + \sum_{i=1}^{n}\sum_{j\neq i}\mathbb{E}_{xy}[m(y^{(i)})m(y^{(j)})]\Bigg)$$

Solution (b): third term, cont.

Continuing from the grouping form:

$$
\begin{aligned}
&\tfrac{1}{n^2}\Bigg(\sum_{i=1}^{n}\mathbb{E}_{xy}[m(y^{(i)})^2] \\
&\qquad+\sum_{i=1}^{n}\sum_{j\neq i}\mathbb{E}_{xy}[m(y^{(i)})m(y^{(j)})]\Bigg) \\
&= \tfrac{1}{n^2}\Bigg(\sum_{i=1}^{n}\mathbb{E}_{xy}[m(y^{(i)})^2] \\
&\qquad+\sum_{i=1}^{n}\sum_{j\neq i}\mathbb{E}_{xy}[m(y^{(i)})]\,\mathbb{E}_{xy}[m(y^{(j)})]\Bigg)
\end{aligned}
$$

Cross term: independence trick.
For $i \neq j$, $m(y^{(i)})$ and $m(y^{(j)})$ are *independent* RVs, so

$$\mathbb{E}_{xy}[m(y^{(i)})m(y^{(j)})] = \mathbb{E}_{xy}[m(y^{(i)})]\,\mathbb{E}_{xy}[m(y^{(j)})].$$

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Continuing from the grouping form:

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= &\tfrac{1}{n^2}\big(n\mathbb{E}_{xy}[m(y)^2] + n(n-1)\mathbb{E}_{xy}[m(y)]^2\big)
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Collapse by i.i.d.

- n diagonal terms, each $= \mathbb{E}_{xy}[m(y)^2]$
  $\Rightarrow\ n\mathbb{E}_{xy}[m(y)^2]$.
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&= \tfrac{1}{n^2}\big(n\mathbb{E}_{xy}[m(y)^2] + n(n-1)\mathbb{E}_{xy}[m(y)]^2\big) \\
&= \tfrac{1}{n}\mathbb{E}_{xy}[m(y)^2] + \big(1-\tfrac{1}{n}\big)\mathbb{E}_{xy}[m(y)]^2
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Solution to Question (b): Continued

Combining the results so far, we get

$$\mathcal{R}_L(\hat{f}) = \mathbb{E}_{xy}[m(y)^2] - 2 \cdot \mathbb{E}_{xy}\left[m(y) \cdot \frac{1}{n}\sum_{i=1}^{n} m(y^{(i)})\right] + \mathbb{E}_{xy}\left[\left(\frac{1}{n}\sum_{i=1}^{n} m(y^{(i)})\right)\left(\frac{1}{n}\sum_{i=1}^{n} m(y^{(i)})\right)\right]$$

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&= \left(1 + \frac{1}{n}\right)\left(\mathbb{E}_{xy}[m(y)^2] - \mathbb{E}_{xy}[m(y)]^2\right)
\end{aligned}
$$

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Combining the results so far, we get

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&= \left(1+\frac{1}{n}\right)\left(\mathbb{E}_{xy}[m(y)^2] - \mathbb{E}_{xy}[m(y)]^2\right) \\
&= \left(1+\frac{1}{n}\right)\mathrm{Var}(m(y)).
\end{aligned}$$

Why does $\mathbb{E}_x$ vanish?

- Constant model $\Rightarrow$ integrands depend on y only.
- Marginalize: $\int p(x,y)\,dx = p(y)$.
- Hence $\mathbb{E}_{xy}[m(y)] = \mathbb{E}_y[m(y)]$; $\mathrm{Var}(m(y))$ is over y.

Question (c)

Derive that the risk minimizer f^* is given by $f^*(\mathbf{x}) = m^{-1}(\mathbb{E}_{y|\mathbf{x}}[m(y)|\mathbf{x}])$.

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Q1: Which hypothesis space?

What $\mathcal{H}$ should we consider when deriving f^*?

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The **unrestricted** space of all functions: $\mathcal{H} = \{f : \mathcal{X} \to \mathbb{R}\}$.

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Q2: What does "unrestricted" imply? Why?

What freedom does this give, and why does it matter?

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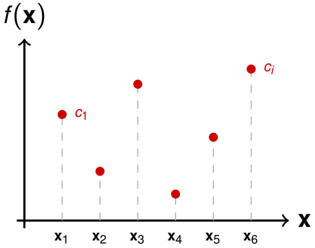
A2

For **each** $\mathbf{x}$, $f(\mathbf{x})$ can take any $c \in \mathbb{R}$ *independently*. $\rightsquigarrow$
Point-wise: minimize risk separately at every $\mathbf{x}$.



Each $\mathbf{x}_i$ gets its own optimal c_i
— no shared functional form.

Solution to Question (c)

Recall (c): point-wise optimization ⇒ decompose

$$\mathbb{E}_{xy}[\,\cdot\,] = \mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{y|\mathbf{x}}[\,\cdot \mid \mathbf{x}]\right].$$

Apply the law of total expectation to the risk:

$$\begin{aligned}\mathcal{R}_L(f) &= \mathbb{E}_{xy}[L(y, f(\mathbf{x}))] \\ &= \mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{y|\mathbf{x}}[(m(y) - m(f(\mathbf{x})))^2 \mid \mathbf{x}]\right].\end{aligned}$$

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Empirical or theoretical risk?

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Which hypothesis space $\mathcal{H}$?

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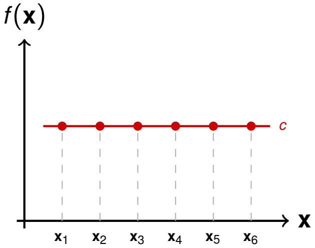
Q2

Empirical or theoretical risk?

A2

The **theoretical** risk $\mathcal{R}_L(f) = \mathbb{E}_{xy}[L(y, f(\mathbf{x}))]$ (not empirical as in (a)).



One c for **all** $\mathbf{x}$.

Solution to Question (d)

The (theoretical) risk for a constant model $\bar{f}(\mathbf{x}) = c$ is:

$$\begin{aligned}
\mathcal{R}_L(\bar{f}) &= \mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{y|\mathbf{x}}\left[(m(y) - m(\bar{f}(\mathbf{x})))^2\right]\right] \\
&= \int_y \int_{\mathbf{x}} (m(y) - m(\bar{f}(\mathbf{x})))^2 p(\mathbf{x}, y) \mathrm{d}\mathbf{x}\mathrm{d}y \\
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Therefore, the optimal constant model is

$$\bar{f}(\mathbf{x}) = c = m^{-1}(\mathbb{E}_y[m(y)])$$

Solution to Question (d): Continued

The **risk** of the optimal constant model $\bar{f}(\mathbf{x}) = c = m^{-1}(\mathbb{E}_y[m(y)])$:

$$\begin{aligned}\mathcal{R}_L(\bar{f}) &= \mathbb{E}_{xy}[(m(y) - m(\bar{f}(\mathbf{x}))^2] \\ &= \mathbb{E}_y[(m(y) - \mathbb{E}_y[m(y)])^2] \\ &= \mathrm{Var}(m(y)).\end{aligned}$$

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Definition of variance: for a random variable Z, $\mathrm{Var}(Z) = \mathbb{E}[(Z - \mathbb{E}[Z])^2]$. Here $Z = m(y)$.

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(e): Recall the decomposition of the Bayes regret into the estimation and the approximation error. Show that the former is $\frac{1}{n}\mathrm{Var}(m(y))$, while the latter is $\mathrm{Var}\left(\mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}]\right)$ for the optimal constant model $\hat{f}(\mathbf{x})$ if $\mathcal{H}$ consists of the constant models.

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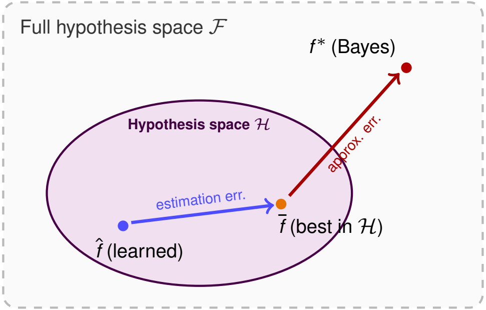
Q1: What is the Bayes regret, and what are its two errors?

A1: Bayes regret $= \mathcal{R}_L(\hat{f}) - \mathcal{R}_L^*$, splits as:

Estimation error:
$\mathcal{R}_L(\hat{f}) - \inf_{f\in\mathcal{H}} \mathcal{R}_L(f)$

Approximation error:
$\inf_{f\in\mathcal{H}} \mathcal{R}_L(f) - \mathcal{R}_L^*$



Solution to Question (e): Roadmap

Three risk minimizers from earlier parts, connected by the two errors:

From (a)

$$\hat{f}(\mathbf{x}) = m^{-1}\left(\frac{1}{n}\sum_{i=1}^{n} m(y^{(i)})\right)$$

$$\mathcal{R}_L(\hat{f}) = (1+\tfrac{1}{n})\mathrm{Var}(m(y))$$

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$\xrightarrow[\text{err.}]{\text{estimation}}$

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$\xrightarrow[\text{err.}]{\text{approximation}}$

From (c)
$f^*(\mathbf{x}) = m^{-1}(\mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}])$
unrestricted min. of $\mathcal{R}_L$

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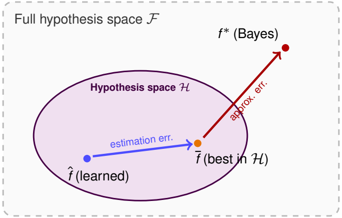

Solution (e): Part 1 of 2 — Estimation Error

$$\mathcal{R}_L(\hat{f}) - \inf_{f \in \mathcal{H}} \mathcal{R}_L(f)$$

$$= \mathcal{R}_L(\hat{f}) - \mathcal{R}_L(\bar{f})$$

$$= \left(1 + \tfrac{1}{n}\right) \mathrm{Var}(m(y)) - \mathrm{Var}(m(y))$$

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$\inf_{f \in \mathcal{H}} \mathcal{R}_L(f) = \mathcal{R}_L(\bar{f})$, since $\bar{f}$ is the in-class minimizer (from (d)).

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Solution (e): Part 2 of 2 — Approximation Error

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&\inf_{f \in \mathcal{H}} \mathcal{R}_L(f) - \mathcal{R}_L^* \\
&= \mathcal{R}_L(\bar{f}) - \mathcal{R}_L(f^*) \\
&= \mathrm{Var}(m(y)) - \mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{y|\mathbf{x}}[(m(y) - m(f^*(\mathbf{x})))^2 \mid \mathbf{x}]\right] \\
&= \mathrm{Var}(m(y)) - \mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{y|\mathbf{x}}[(m(y) - \mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}])^2 \mid \mathbf{x}]\right] \\
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From (d): $\mathcal{R}_L(\bar{f}) = \mathrm{Var}(m(y))$.
Def.: $\mathcal{R}_L(f^*) = \mathbb{E}_{\mathbf{x}}[\mathbb{E}_{y|\mathbf{x}}[(m(y) - m(f^*))^2 \mid \mathbf{x}]]$.

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&= \mathcal{R}_L(\bar{f}) - \mathcal{R}_L(f^*) \\
&= \mathrm{Var}(m(y)) - \mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{y|\mathbf{x}}[(m(y) - m(f^*(\mathbf{x})))^2 \mid \mathbf{x}]\right] \\
&= \mathrm{Var}(m(y)) - \mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{y|\mathbf{x}}[(m(y) - \mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}])^2 \mid \mathbf{x}]\right] \\
&= \mathrm{Var}(m(y)) - \mathbb{E}_{\mathbf{x}}[\mathrm{Var}(m(y) \mid \mathbf{x})] \\
&= \boxed{\mathrm{Var}(\mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}])}
\end{aligned}
$$

From (d): $\mathcal{R}_L(\bar{f}) = \mathrm{Var}(m(y))$.
Def.: $\mathcal{R}_L(f^*) = \mathbb{E}_{\mathbf{x}}[\mathbb{E}_{y|\mathbf{x}}[(m(y) - m(f^*))^2 \mid \mathbf{x}]]$.

From (c): $f^*(\mathbf{x}) = m^{-1}(\mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}])$, so $m(f^*(\mathbf{x})) = \mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}]$.

Solution (e): Part 2 of 2 — Approximation Error

$$\begin{aligned}
&\inf_{f\in\mathcal{H}} \mathcal{R}_L(f) - \mathcal{R}_L^* \\
&= \mathcal{R}_L(\bar{f}) - \mathcal{R}_L(f^*) \\
&= \mathrm{Var}(m(y)) - \mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{y|\mathbf{x}}[(m(y) - m(f^*(\mathbf{x})))^2 \mid \mathbf{x}]\right] \\
&= \mathrm{Var}(m(y)) - \mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{y|\mathbf{x}}[(m(y) - \mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}])^2 \mid \mathbf{x}]\right] \\
&= \mathrm{Var}(m(y)) - \mathbb{E}_{\mathbf{x}}[\mathrm{Var}(m(y) \mid \mathbf{x})] \\
&= \boxed{\mathrm{Var}(\mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}])}
\end{aligned}$$

From (d): $\mathcal{R}_L(\bar{f}) = \mathrm{Var}(m(y))$.
Def.: $\mathcal{R}_L(f^*) = \mathbb{E}_{\mathbf{x}}[\mathbb{E}_{y|\mathbf{x}}[(m(y) - m(f^*))^2 \mid \mathbf{x}]]$.

From (c): $f^*(\mathbf{x}) = m^{-1}(\mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}])$, so $m(f^*(\mathbf{x})) = \mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}]$.

Solution (e): Part 2 of 2 — Approximation Error

$$
\begin{aligned}
&\inf_{f \in \mathcal{H}} \mathcal{R}_L(f) - \mathcal{R}_L^* \\
&= \mathcal{R}_L(\bar{f}) - \mathcal{R}_L(f^*) \\
&= \mathrm{Var}(m(y)) - \mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{y|\mathbf{x}}[(m(y) - m(f^*(\mathbf{x})))^2 \mid \mathbf{x}]\right] \\
&= \mathrm{Var}(m(y)) - \mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{y|\mathbf{x}}[(m(y) - \mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}])^2 \mid \mathbf{x}]\right] \\
&= \mathrm{Var}(m(y)) - \mathbb{E}_{\mathbf{x}}[\mathrm{Var}(m(y) \mid \mathbf{x})] \\
&= \boxed{\mathrm{Var}(\mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}])}
\end{aligned}
$$

From (d): $\mathcal{R}_L(\bar{f}) = \mathrm{Var}(m(y))$.
Def.: $\mathcal{R}_L(f^*) = \mathbb{E}_{\mathbf{x}}[\mathbb{E}_{y|\mathbf{x}}[(m(y) - m(f^*))^2 \mid \mathbf{x}]]$.

From (c): $f^*(\mathbf{x}) = m^{-1}(\mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}])$, so $m(f^*(\mathbf{x})) = \mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}]$.

Law of total variance: $\mathrm{Var}(Y) = \mathbb{E}_X[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(\mathbb{E}[Y \mid X])$.